Table 6. In-distribution (ID) and out-of-distribution (OOD) evaluation results. For a fair comparison, all methods are trained with the same total number of rollouts. Evaluation is conducted over 4 independent runs, with Acc@8 computed using 8 samples per run (i.e., 32 responses per prompt in total). Each value reports the mean $\pm$ standard deviation across the 4 runs. Δ rows indicate the improvement of GRPO-DBB over each baseline, with one-sided paired t-test p-values (H_1: GRPO-DBB $>$ baseline).

	In-Distribution							**Out-of-Distribution**			
Method	MATH500	Minerva	AIME24	AIME25	AMC24	Olympiad	Avg.	MMLU-Pro	GPQA-D	BBH	Avg.
				Qwen3-8B-Base trained with **DAPO-Math-17k**							
GRPO	87.87 $\pm$ 0.36	39.34 $\pm$ 0.29	29.17 $\pm$ 0.68	25.31 $\pm$ 2.16	57.71 $\pm$ 1.44	54.54 $\pm$ 0.19	48.99 $\pm$ 0.46	50.16 $\pm$ 0.33	45.27 $\pm$ 1.69	59.18 $\pm$ 0.33	51.54 $\pm$ 0.55
RePO	86.22 $\pm$ 0.31	35.02 $\pm$ 0.47	27.29 $\pm$ 0.80	23.44 $\pm$ 1.04	56.25 $\pm$ 0.70	52.70 $\pm$ 0.44	46.82 $\pm$ 0.07	53.29 $\pm$ 0.55	46.34 $\pm$ 1.74	42.44 $\pm$ 0.30	47.35 $\pm$ 0.80
GRESO	87.01 $\pm$ 0.20	33.88 $\pm$ 0.60	27.36 $\pm$ 0.79	22.78 $\pm$ 1.09	57.31 $\pm$ 0.57	52.81 $\pm$ 0.17	46.86 $\pm$ 0.12	62.81 $\pm$ 0.39	48.53 $\pm$ 0.43	65.17 $\pm$ 0.63	58.83 $\pm$ 0.40
DAPO	86.00 $\pm$ 0.07	37.08 $\pm$ 0.30	31.06 $\pm$ 1.19	21.02 $\pm$ 1.14	52.67 $\pm$ 0.92	51.64 $\pm$ 0.14	46.58 $\pm$ 0.16	59.28 $\pm$ 0.67	45.05 $\pm$ 1.79	63.41 $\pm$ 0.20	55.91 $\pm$ 0.46
GRPO-DBB	**88.71 $\pm$ 0.18**	**39.46 $\pm$ 0.51**	**33.60 $\pm$ 2.05**	**29.02 $\pm$ 2.53**	**60.54 $\pm$ 1.00**	**56.53 $\pm$ 0.25**	**51.31 $\pm$ 0.53**	**63.12 $\pm$ 0.38**	**49.43 $\pm$ 2.11**	**67.09 $\pm$ 0.13**	**59.88 $\pm$ 0.66**
Δ w.r.t. GRPO	+0.84 $\pm$ 0.33 (p=0.0073)	+0.12 $\pm$ 0.24 (p=0.2015)	+4.44 $\pm$ 2.05 (p=0.0114)	+3.71 $\pm$ 1.38 (p=0.0063)	+2.83 $\pm$ 0.68 (p=0.0018)	+1.99 $\pm$ 0.37 (p=0.0009)	+2.32 $\pm$ 0.41 (p=0.0007)	+12.96 $\pm$ 0.53 (p=0.0000)	+4.17 $\pm$ 1.46 (p=0.0053)	+7.91 $\pm$ 0.38 (p=0.0000)	+8.34 $\pm$ 0.30 (p=0.0000)
Δ w.r.t. RePO	+2.49 $\pm$ 0.44 (p=0.0008)	+4.44 $\pm$ 0.23 (p=0.0000)	+6.31 $\pm$ 2.46 (p=0.0072)	+5.58 $\pm$ 3.42 (p=0.0234)	+4.29 $\pm$ 1.33 (p=0.0038)	+3.83 $\pm$ 0.35 (p=0.0001)	+4.49 $\pm$ 0.50 (p=0.0002)	+9.84 $\pm$ 0.68 (p=0.0000)	+3.09 $\pm$ 0.82 (p=0.0024)	+24.65 $\pm$ 0.39 (p=0.0000)	+12.53 $\pm$ 0.40 (p=0.0000)
Δ w.r.t. GRESO	+1.70 $\pm$ 0.34 (p=0.0011)	+5.57 $\pm$ 0.93 (p=0.0006)	+6.24 $\pm$ 2.49 (p=0.0077)	+6.24 $\pm$ 2.17 (p=0.0052)	+3.23 $\pm$ 1.42 (p=0.0100)	+3.72 $\pm$ 0.39 (p=0.0002)	+4.45 $\pm$ 0.50 (p=0.0002)	+0.32 $\pm$ 0.09 (p=0.0032)	+0.90 $\pm$ 1.68 (p=0.1806)	+1.92 $\pm$ 0.64 (p=0.0045)	+1.05 $\pm$ 0.42 (p=0.0077)
Δ w.r.t. DAPO	+2.71 $\pm$ 0.17 (p=0.0000)	+2.37 $\pm$ 0.49 (p=0.0011)	+2.54 $\pm$ 1.52 (p=0.0220)	+8.00 $\pm$ 1.58 (p=0.0010)	+7.87 $\pm$ 1.63 (p=0.0012)	+4.89 $\pm$ 0.33 (p=0.0000)	+4.73 $\pm$ 0.47 (p=0.0001)	+3.84 $\pm$ 0.82 (p=0.0013)	+4.39 $\pm$ 1.07 (p=0.0019)	+3.68 $\pm$ 0.31 (p=0.0001)	+3.97 $\pm$ 0.28 (p=0.0001)
				Qwen3-1.7B-Base trained with **DAPO-Math-17k**							
GRPO	67.50 $\pm$ 0.79	25.29 $\pm$ 0.31	9.06 $\pm$ 0.63	4.48 $\pm$ 0.21	23.40 $\pm$ 1.39	29.57 $\pm$ 0.37	26.55 $\pm$ 0.26	33.62 $\pm$ 0.09	21.90 $\pm$ 1.04	12.71 $\pm$ 0.17	22.74 $\pm$ 0.32
RePO	67.27 $\pm$ 0.55	23.78 $\pm$ 0.12	7.60 $\pm$ 1.50	5.94 $\pm$ 0.71	22.57 $\pm$ 1.39	29.00 $\pm$ 0.43	26.03 $\pm$ 0.29	36.58 $\pm$ 0.73	25.25 $\pm$ 0.99	**30.37 $\pm$ 0.41**	30.73 $\pm$ 0.32
GRESO	67.60 $\pm$ 0.05	23.76 $\pm$ 0.20	8.19 $\pm$ 1.37	6.39 $\pm$ 0.79	23.61 $\pm$ 1.96	30.91 $\pm$ 0.05	26.74 $\pm$ 0.09	35.68 $\pm$ 0.35	27.15 $\pm$ 1.24	26.67 $\pm$ 0.18	29.83 $\pm$ 0.46
DAPO	66.69 $\pm$ 0.13	24.34 $\pm$ 0.23	9.72 $\pm$ 0.86	5.42 $\pm$ 0.90	19.72 $\pm$ 0.45	28.70 $\pm$ 0.06	25.77 $\pm$ 0.33	33.71 $\pm$ 0.69	24.49 $\pm$ 0.86	21.05 $\pm$ 0.12	26.42 $\pm$ 0.42
GRPO-DBB	**71.67 $\pm$ 0.22**	**26.43 $\pm$ 0.67**	**13.75 $\pm$ 1.13**	**7.40 $\pm$ 1.04**	**27.31 $\pm$ 1.90**	**33.36 $\pm$ 0.50**	**29.98 $\pm$ 0.28**	**40.81 $\pm$ 0.11**	**27.89 $\pm$ 1.46**	29.67 $\pm$ 0.89	**32.79 $\pm$ 0.79**
Δ w.r.t. GRPO	+4.17 $\pm$ 0.60 (p=0.0004)	+1.14 $\pm$ 0.89 (p=0.0419)	+4.69 $\pm$ 1.75 (p=0.0063)	+2.92 $\pm$ 0.90 (p=0.0037)	+3.90 $\pm$ 2.05 (p=0.0158)	+3.79 $\pm$ 0.66 (p=0.0007)	+3.43 $\pm$ 0.27 (p=0.0001)	+7.19 $\pm$ 0.10 (p=0.0000)	+5.98 $\pm$ 2.48 (p=0.0085)	+16.97 $\pm$ 0.90 (p=0.0000)	+10.05 $\pm$ 1.10 (p=0.0002)
Δ w.r.t. RePO	+4.40 $\pm$ 0.52 (p=0.0002)	+2.64 $\pm$ 0.59 (p=0.0015)	+6.14 $\pm$ 1.04 (p=0.0007)	+1.46 $\pm$ 0.87 (p=0.0219)	+4.74 $\pm$ 2.73 (p=0.0202)	+4.35 $\pm$ 0.60 (p=0.0004)	+3.95 $\pm$ 0.43 (p=0.0002)	+4.24 $\pm$ 0.74 (p=0.0007)	+2.64 $\pm$ 1.20 (p=0.0110)	-0.70 $\pm$ 1.25 (p=0.8273)	+2.06 $\pm$ 0.35 (p=0.0007)
Δ w.r.t. GRESO	+4.07 $\pm$ 0.20 (p=0.0000)	+2.67 $\pm$ 0.83 (p=0.0039)	+5.56 $\pm$ 0.44 (p=0.0001)	+1.01 $\pm$ 0.56 (p=0.0184)	+3.69 $\pm$ 2.16 (p=0.0209)	+2.44 $\pm$ 0.47 (p=0.0010)	+3.24 $\pm$ 0.29 (p=0.0001)	+5.13 $\pm$ 0.39 (p=0.0001)	+0.74 $\pm$ 0.54 (p=0.0354)	+3.01 $\pm$ 0.94 (p=0.0038)	+2.96 $\pm$ 0.25 (p=0.0001)
Δ w.r.t. DAPO	+4.98 $\pm$ 0.24 (p=0.0000)	+2.08 $\pm$ 0.60 (p=0.0031)	+4.03 $\pm$ 1.89 (p=0.0119)	+1.98 $\pm$ 1.50 (p=0.0387)	+7.58 $\pm$ 1.82 (p=0.0018)	+4.66 $\pm$ 0.45 (p=0.0001)	+4.22 $\pm$ 0.47 (p=0.0002)	+7.11 $\pm$ 0.79 (p=0.0002)	+3.39 $\pm$ 1.59 (p=0.0119)	+8.62 $\pm$ 0.88 (p=0.0001)	+6.38 $\pm$ 0.92 (p=0.0004)